
MULTI-AGENT DUE-DILIGENCE FOR COMMERCIAL REAL ESTATE

The institutional diligence committee, run by **agents**.

*eight separate agents,
one shared room*

Eight specialist agents work inside a single Band room. They read each other's findings, settle the conflicts between them, hand each other work, and call in quantitative models when a deal needs them. The result is a deal memo an investment committee can sign off on.

— THE PROBLEM

Diligence is run by a committee of experts. The committee is where deals slow down.

Buying one asset means title, zoning, environmental, structural, insurance and financial review. Those experts work in sequence over email and PDFs, and an associate assembles their findings by hand before the deal can close.

SLOW & SERIAL

3–6 wks

to clear diligence on a mid-market asset, most of it lost to hand-offs between specialists.

COSTLY TO GET WRONG

\$100M+

a single missed easement, zoning conflict or remediation liability can move an IRR below water.

NO AUDIT TRAIL

Opaque

the rationale for an approval lives across inboxes — difficult to replay or defend to an IC.

Our read

The committee can run in minutes if every expert shares one context, the conflicts between them surface on their own, and each finding and number is recorded as it happens.

— THE SOLUTION

A standing committee of agents that coordinate through Band.

- 01 **Read the room.** Each agent pulls the live Band conversation as its primary context before it reasons.
- 02 **Challenge.** When two agents disagree — an easement one finds and another misses — they resolve it in-thread.
- 03 **Delegate.** Any agent can assign another a defined task, with intent and authority, derived from context.
- 04 **Recruit.** The committee brings Python quantitative specialists into the room when the deal warrants it.
- 05 **Synthesize.** A Deal Director composes a Red / Yellow / Green memo with conditions precedent.

OUTCOME

A deal memo a committee can defend, with the full reasoning attached, in minutes.

DECIDED AT RUNTIME

Which agents debate, who hands work to whom, and which specialists join are **read from the deal**, not fixed in code. The collaboration is emergent.

Eight agents. Two frameworks. One room.

● TypeScript · reasoning

● Python · quantitative

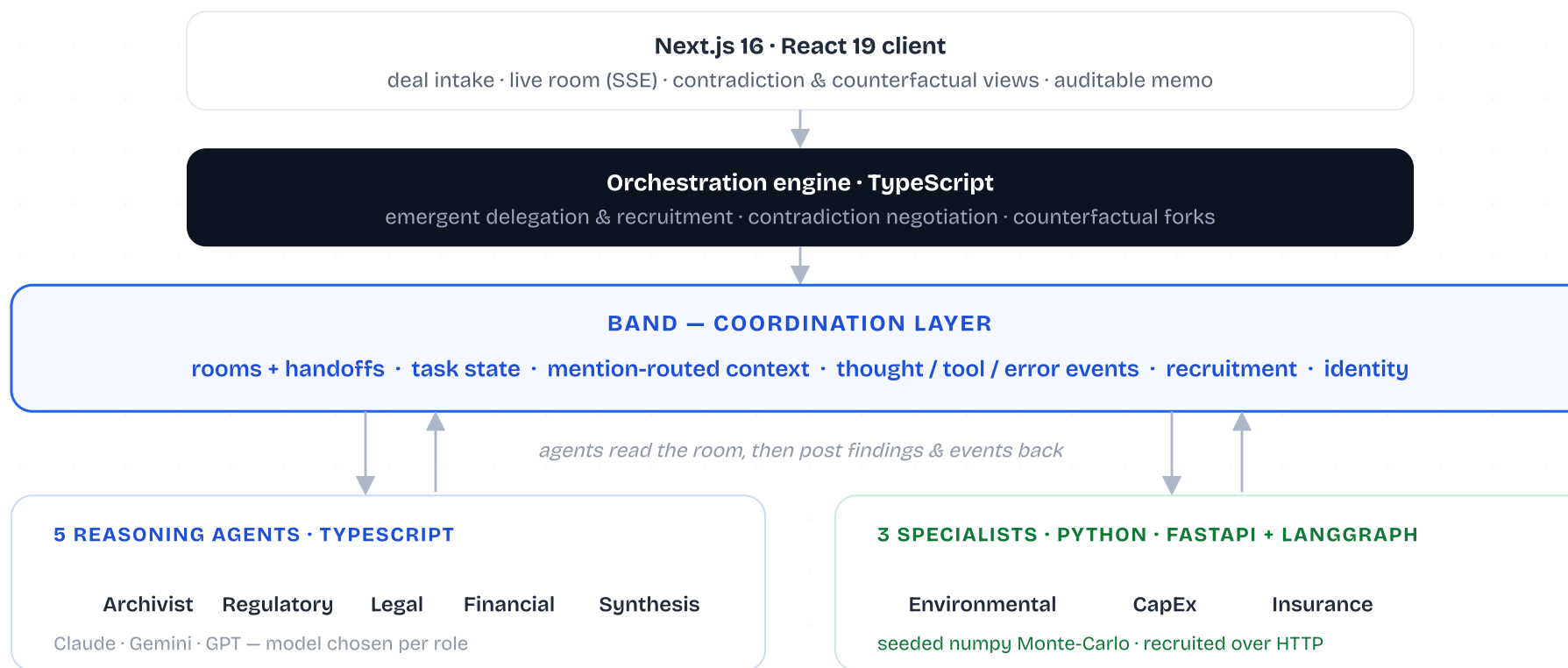
● Archivist Document intake → structured property facts and missing-document gaps.	● Regulatory Zoning, permits, FEMA flood and deed restrictions, severity-ranked.	● Legal Risk Title defects, easement conflicts, liens and seller-representation gaps.	● Financial NOI, cap rate, DSCR and a 5-year IRR — computed deterministically.
● Synthesis Deal Director — composes the R / Y / G memo and conditions precedent.	● Environmental Contamination risk and Monte-Carlo remediation cost; Phase I call.	● CapEx / Construction Renovation and conversion cost with schedule risk, simulated.	● Insurance / Catastrophe Flood, wind and seismic exposure with a premium estimate.

Why two frameworks?

The reasoning agents live in TypeScript alongside the product. The specialists are Python because they perform genuine **numerical work** — seeded numpy Monte-Carlo over LangGraph state machines. Band unifies them into one committee.

Band is where the agents meet.

The orchestrator decides the order of work. The agents themselves talk through Band: they read the shared context, post their findings, and emit their reasoning as events.



One deal, start to finish.

Six stages run in order. What happens inside each one (which agents debate, who delegates, which specialists join) is read from the deal.



EMITTED TO BAND THROUGHOUT

Every stage posts `thought`, `tool_call`, `tool_result` and `error` events, and delegations are tracked as Band tasks — the whole run replays from the record.

AFTER THE RUN — OPTIONAL

The partner can fork a counterfactual (“drop the as-of-right use”) and re-run the committee to compare.

What the committee relies on Band for.

Take any of these away and the committee stops working. We use the full Band surface, not just messaging.

Shared context Agents read the room via mention-routed getContext before reasoning — primary context, not spoon-fed payloads.	Handoffs Structured, mention-based posting passes the baton along the committee, each agent addressing the next.	Task state Delegations are real Band tasks tracked processing → processed — accountable, not fire-and-forget.
Rich events thought · tool_call · tool_result · error make each agent's reasoning visible and replayable.	Recruitment Specialists are pulled into the live room mid-deal — cross-framework agents joining as first-class participants.	Distinct identity Each specialist posts under its own Band identity (per-agent key) — genuinely separate agents.

The agents decide the next step.

lib/agents/schemas.ts

```
// each agent emits these from the deal context
export const ComplianceReportSchema = z.object({
  // ...findings, risk_score...
  delegations: z.array(DelegationRequestSchema),
  requested_specialists: z.array(SpecialistRequestSchema),
});

const DelegationRequestSchema = z.object({
  to: z.enum(['archivist', 'regulatory', 'legal',
    'financial', 'synthesis']),
  intent: z.string(), // what + why
  authority: z.string(), // what may change
});
```

lib/orchestration/workflow.ts

```
// the agents decide who does what next
for (const d of delegations) {
  await delegate(dealId, roomId, d.from, d.to,
    d.intent, d.authority, async () =>
      run(d.to, { ...ctx, delegation: d }));
  // → real Band task: processing → processed
}
```

WHAT THIS ENABLES

Delegations and specialist requests are values the agents produce, not control flow we wrote. A zoning conflict routes Legal to Financial; a contamination flag pulls in Environmental. Both come from the deal.

Watch the committee work, in real time.

● Cedar Wharf · diligence room

8 agents · live

■ Archivist → Regulatory

Extracted property facts. Flagging two missing documents: ALTA survey, Phase I.

■ Legal Risk

Contract references a neighbour access easement — **conflicts** with the title record.

Contradiction · Archivist vs Legal — negotiating...

● Environmental recruited by Regulatory

Phase I warranted. Remediation P50 ≈ \$2.4M.

task Legal → Financial: re-underwrite · **processed** ✓

DEAL MEMO

YELLOW

Proceed with three conditions precedent: resolve the easement, complete Phase I, and re-price to a **9.1% IRR** floor.

LIVE EVENT STREAM

```
thought regulatory: assessing flood zone...
tool_call environmental.simulate(trials=1e4)
tool_result P50 2.4M · P90 6.1M
handoff legal → financial
financial.recalculated 14.2% → 9.1%
```

Python specialists that do real quantitative work.

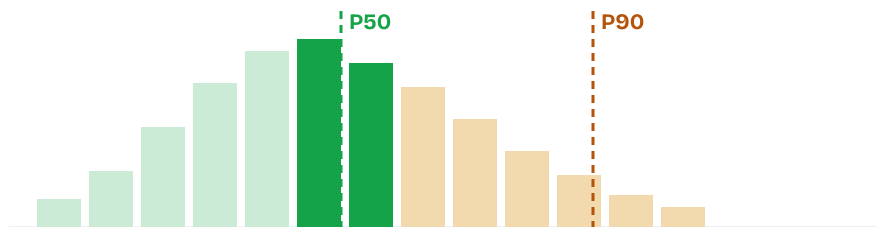
Three LangGraph state machines run seeded numpy Monte-Carlo — quantitative analysis the reasoning agents cannot fabricate. They are recruited live and post results back to the room.

CAPEX SPECIALIST — SAMPLE RUN · CEDAR WHARF

\$186M

P50 conversion cost

100% over budget at P50



ENVIRONMENTAL

Contamination-risk score and a remediation-cost distribution → a defensible **Phase I** recommendation.

INSURANCE / CATASTROPHE

Flood, wind and seismic event simulation → expected annual loss and a grounded premium estimate.

WHY PYTHON

The framework is not the point — the value is. numpy and LangGraph are the right tools for stochastic modelling, and Band makes them one committee regardless.

Numbers you can defend to an IC.

DETERMINISTIC UNDERWRITING

The IRR is not generated by a language model. The Financial agent's headline number is computed in code — DSCR and a levered 5-year IRR solved by **bisection** — so identical inputs always yield the same answer.

Baseline IRR

14.2%



After zoning cascade

9.1%

A critical upstream flag delegates a re-underwrite, and the IRR moves accordingly.

FULL AUDIT TRAIL

handoff Archivist → Regulatory: facts + 2 missing docs

conflict Legal vs Archivist easement — negotiated, resolved

recruit Regulatory pulled in Environmental specialist

task Legal → Financial: re-underwrite · processed ✓

memo Synthesis: **YELLOW** — 3 conditions precedent

Every line is persisted to Postgres and replayable — the answer to “why did we approve this?”

A large, underserved, high-value market.

TARGET USER

Institutional and mid-market commercial-real-estate acquirers — private-equity real-estate funds, REITs and family offices closing 20–100+ assets a year.

WHY NOW

Reliable multi-agent reasoning and a coordination layer like Band have only just become production-viable. Diligence is the ideal first wedge: structured, repetitive, and expensive to get wrong.

REVENUE MODEL

Enterprise SaaS seats plus per-deal pricing — on the order of a few thousand dollars per completed diligence, against the \$50k+ a single deal spends on outside counsel and consultants.

~\$24B

TAM

Annual US spend on diligence, legal and advisory across commercial-real-estate transactions.

~\$3.5B

SAM

Technology-forward institutional and mid-market acquirers reachable with a software product.

10×

Cost & time advantage

Minutes instead of weeks, at a fraction of per-deal advisory cost — with a defensible audit trail.

Market figures are internal estimates for illustration; the wedge is the per-deal value, not the headline TAM.

One room. A modern, cross-framework stack.



Coordinated by Band — the layer every agent reads from and writes to.

FRONTEND & APP



Next.js



React



TypeScript



Tailwind

PYTHON SPECIALISTS



Python



FastAPI



LangGraph



numpy

AI MODELS



Claude



Gemini



OpenAI

DATA & INFRA



Neon



Postgres



Drizzle



Vercel



Eight agents become one diligence committee, because Band is what coordinates them.

Genuine coordination, quantitative specialists across two frameworks, and underwriting an investment committee can stand behind.

...and nothing hardcoded.